

EE-312L Signals and Systems Lab
Laboratory Session 08
**To Understand and Demonstrate the
Continuous-Time Fourier Series of Periodic
Signals Using MATLAB**

December 4, 2024

1 Objective

The objective of this lab session is to understand and demonstrate the continuous-time Fourier Series using MATLAB.

2 Continuous-Time Fourier Series

For continuous-time signal $x(t)$ such that $x(t) = x(t + T)$, where T is the time period of the signal. For fundamental frequency ω_0 , $T = \frac{2\pi}{\omega_0}$. Harmonically related signals with the complex exponential are given as:

$$\phi_k = e^{jk\omega_0 t} = e^{jk\frac{2\pi}{T}t}, \quad (1)$$

where $k = 0, \pm 1, \pm 2, \pm 3 \dots$

2.1 Synthesis Equation

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{2\pi}{T}t} \quad (2)$$

2.2 Analysis Equation

$$a_k = \frac{1}{T} \int_0^T x(t) e^{-jk\omega_0 t} dt \quad (3)$$

```
1 function [sum, t] = fseries(N, M)
2 %N = Number of points plotted
3 t = linspace(-1,1,N);
4 sum = 0.*t;
5 % M = number of coefficients
6 a=[];
7 for k = 1:2:M
8     sum = sum + 4/pi*sin(k*pi*t)/k;
9     a=[a,4/pi/k];
10
11 end
```

```
1
2 clear all;
3 close all;
4 clc;
5 %N = No. of points
6 %M=No of coefficients
7
8
9 [sum, t] = fseries(1000, 100);
10 f = sign(t);
11
12 figure(1)
13 plot(t, sum, 'r', 'linewidth', 2.5)
14 hold on
15 plot(t,f,'LineWidth',2)
16 hold on
17 error = abs(sum-f);
18 plot(t, error, 'm');
19 legend('sum of sinusoides', 'square signal', 'error')
20 axis([-1 1 -1.5 2])
```

Example 1:

Consider the square wave function

$$x(t) = \begin{cases} -1, & -1 \leq t < 0 \\ 1, & 0 \leq t \leq 1 \end{cases}$$

The Fourier sum of sines can be used to accurately approximate the square wave function. The more points plotted and coefficients used the closer the Fourier sum will be to looking like the square wave function. The equation of Fourier sine series used in this case is represented

as follows:

$$x(t) = \frac{4}{\pi} \sum_{k=odd}^{\infty} \frac{1}{k} \sin(k\pi t) \quad (4)$$

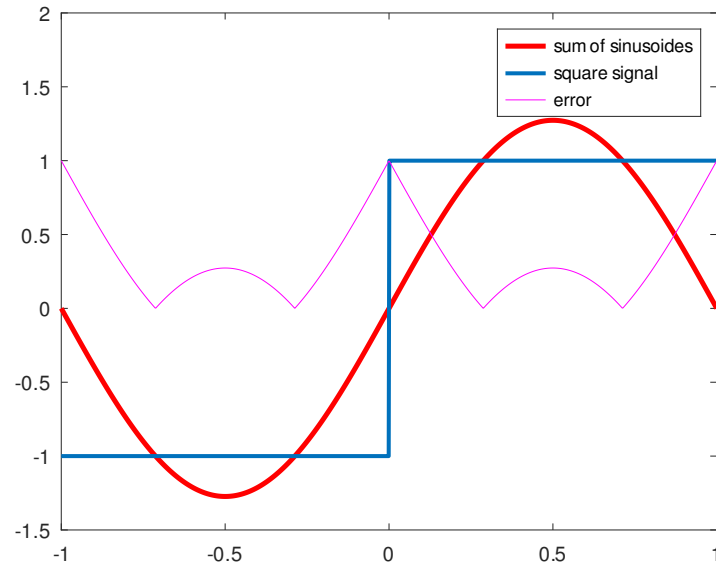
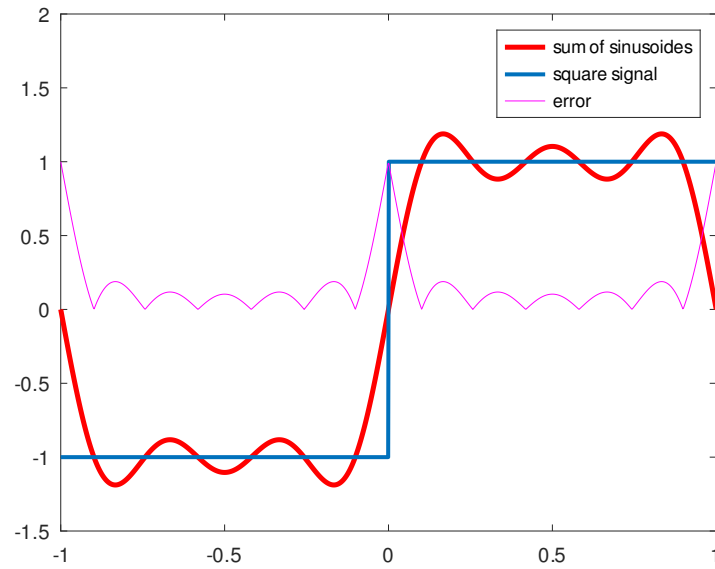


Figure 1: $k=1$

Figure 2: $k=5$

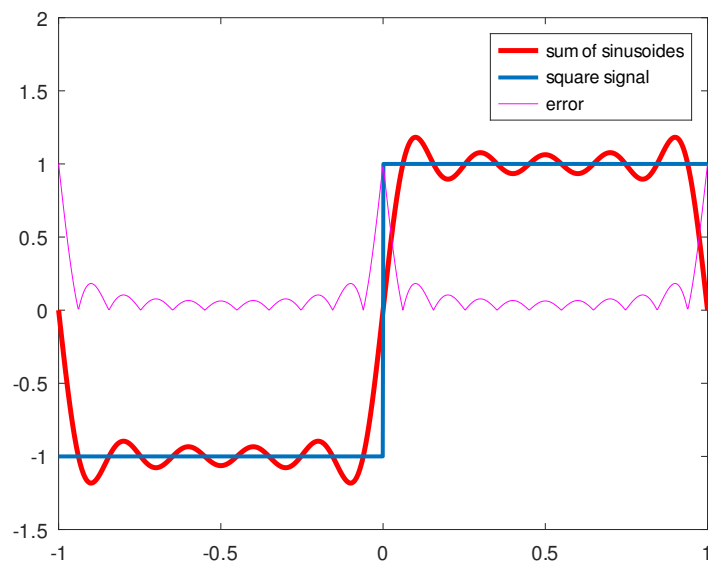
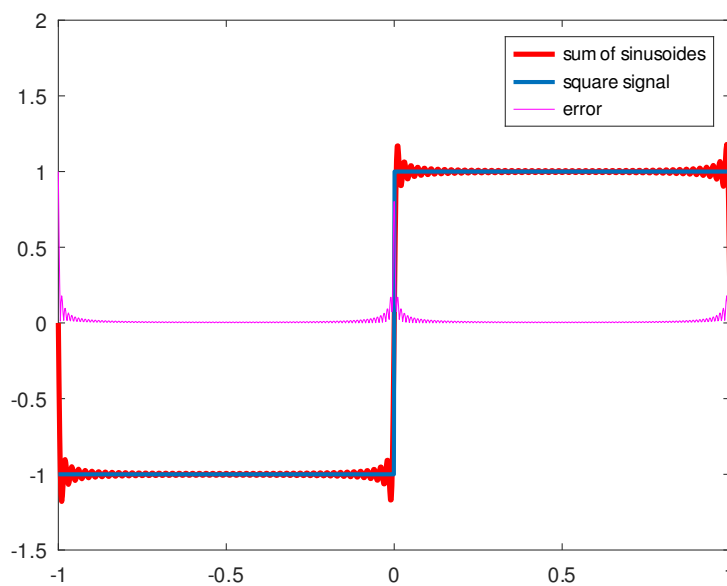
3 Lab 08: Lab Tasks

1. Write a MATLAB program that plots the signal $s(t)$, given as:

$$s(t) = \sum_{k=1}^K \frac{\sin(2\pi kt)}{k}, \quad (5)$$

where $k = 1, 3, 5, 7, 9, 11, 13$, $K = 13$

2. Find the Fourier series expressions for the signals given in Figure 5, manually. Then plot the Fourier series expressions using MATLAB for different values of k .

Figure 3: $k=10$ Figure 4: $k=100$

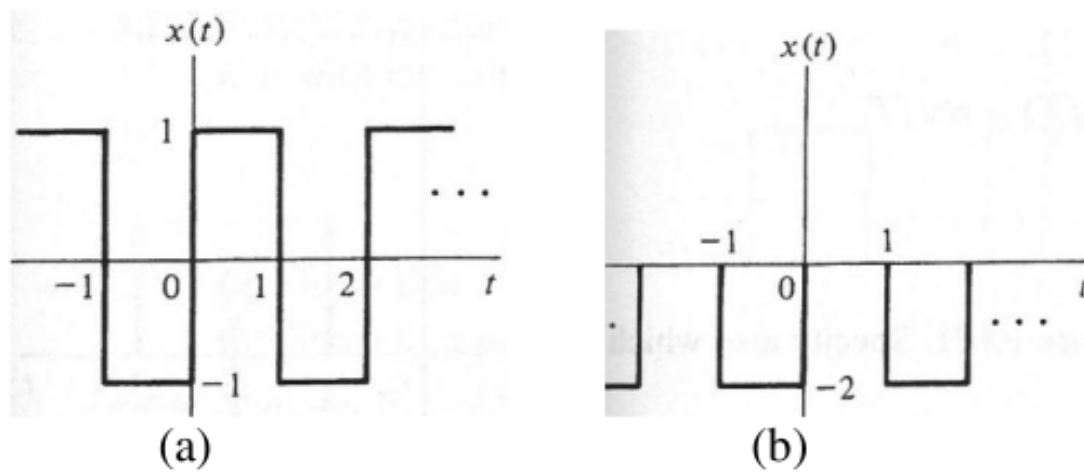


Figure 5: Periodic continuous-time signals for Lab Task.